

Delhi Public School

Subject: Biology | Grade: 12th | Class: 12th

Time: 2 Hours

Maximum Marks: 13

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

Objective Type Questions. Each question carries 1 mark. Choose the correct option.

Q1. Which of the following crosses is used by geneticists to determine the homozygous or heterozygous dominance of a trait in an F1 individual? [Easy] [1]

- a. Self-pollination of F1 generation
- b. Crossing F1 hybrid with the dominant parent
- c. Crossing F1 hybrid with the double recessive parent
- d. Crossing two heterozygous dominant parents

Q2. The AB blood group in humans is a classic example of codominance. This means that in a heterozygous individual with IA IB genotype: [Easy] [1]

- a. Allele IA is partially dominant over IB
- b. Both alleles IA and IB express themselves fully and independently
- c. Allele IB completely masks the expression of IA
- d. A completely new intermediate antigen is formed on RBCs

Q3. In a pea plant, a dihybrid cross is performed between a plant heterozygous for yellow and round seeds (RrYy) and another plant with green and wrinkled seeds (rryy). What is the expected phenotypic ratio of the offspring? [Moderate] [1]

- a. 9:3:3:1
- b. 1:1:1:1
- c. 3:1
- d. 1:2:1

Q4. According to the Hardy-Weinberg principle, if the frequency of a recessive allele 'a' in a stable population is 0.3, what is the expected frequency of the heterozygous genotype 'Aa'? [Moderate] [1]

- a. 0.09
- b. 0.49
- c. 0.42
- d. 0.21

Q5. Industrial melanism in the peppered moth (Biston betularia) in England during the industrial revolution represents a shift from light-colored moths to dark-colored moths. This phenomenon illustrates which type of natural selection? [Hard] [1]

- a. Stabilizing selection
- b. Directional selection
- c. Disruptive selection
- d. Artificial selection

Section B

Short Answer Type Questions. Each question carries 2 marks. Answer in 30-50 words.

Q6. State Mendel's Law of Segregation and explain why it is also known as the 'Law of Purity of Gametes'. [Easy] [2]

Q7. Using a Punnett square, determine the phenotypic and genotypic ratios of the F₂ generation in a cross showing incomplete dominance, such as flower color in Snapdragon (*Antirrhinum majus*) crossed between red (RR) and white (rr) flowered plants. [Moderate] **[2]**

Q8. Differentiate between homologous and analogous organs. How do they provide evidence for divergent and convergent evolution respectively? [Moderate] **[2]**

Section C

Long Answer Type Questions. Each question carries 1 mark. Answer in detail.

Q9. State the Hardy-Weinberg principle. Identify and evaluate any three major evolutionary factors that can upset the genetic equilibrium of a population, leading to evolutionary change. [Moderate] **[1]**

Q10. Imagine you are a plant geneticist. You cross a pure-breeding plant with blue flowers with another pure-breeding plant of the same species with yellow flowers. All the resulting F₁ plants produce green flowers. Formulate a hypothesis to explain this inheritance pattern. Outline a complete genetic cross (including parental, F₁, and expected F₂ phenotypic and genotypic ratios) to support your hypothesis, and design a test cross to confirm the genotype of the F₁ green-flowered plants. [Hard] **[1]**

— End of Question Paper —